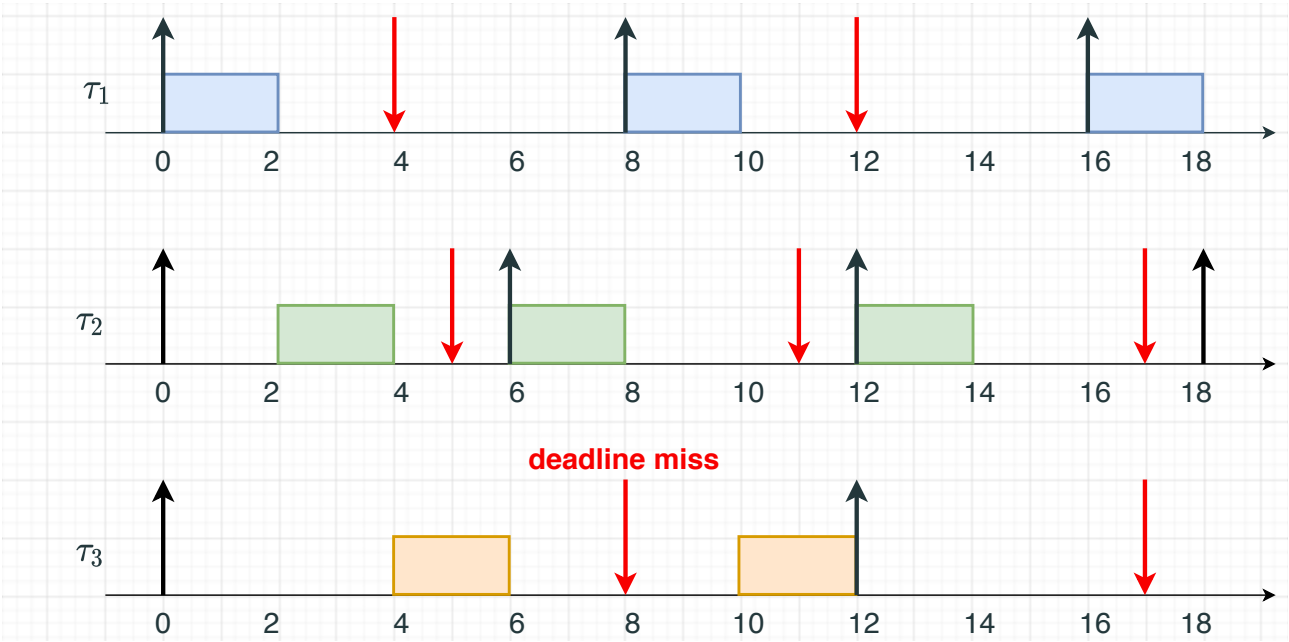
DM: example of an unfeasible schedule (1/2)

PER COSTRUIRE LO SCHEDULE SI PARTE DAL TASK CON PRIORITÀ PIÙ ALTA.

task τ_i	C_i	T_i	D_i
τ_1	2	8	4
τ_2	2	6	5
τ_3	4	12	8

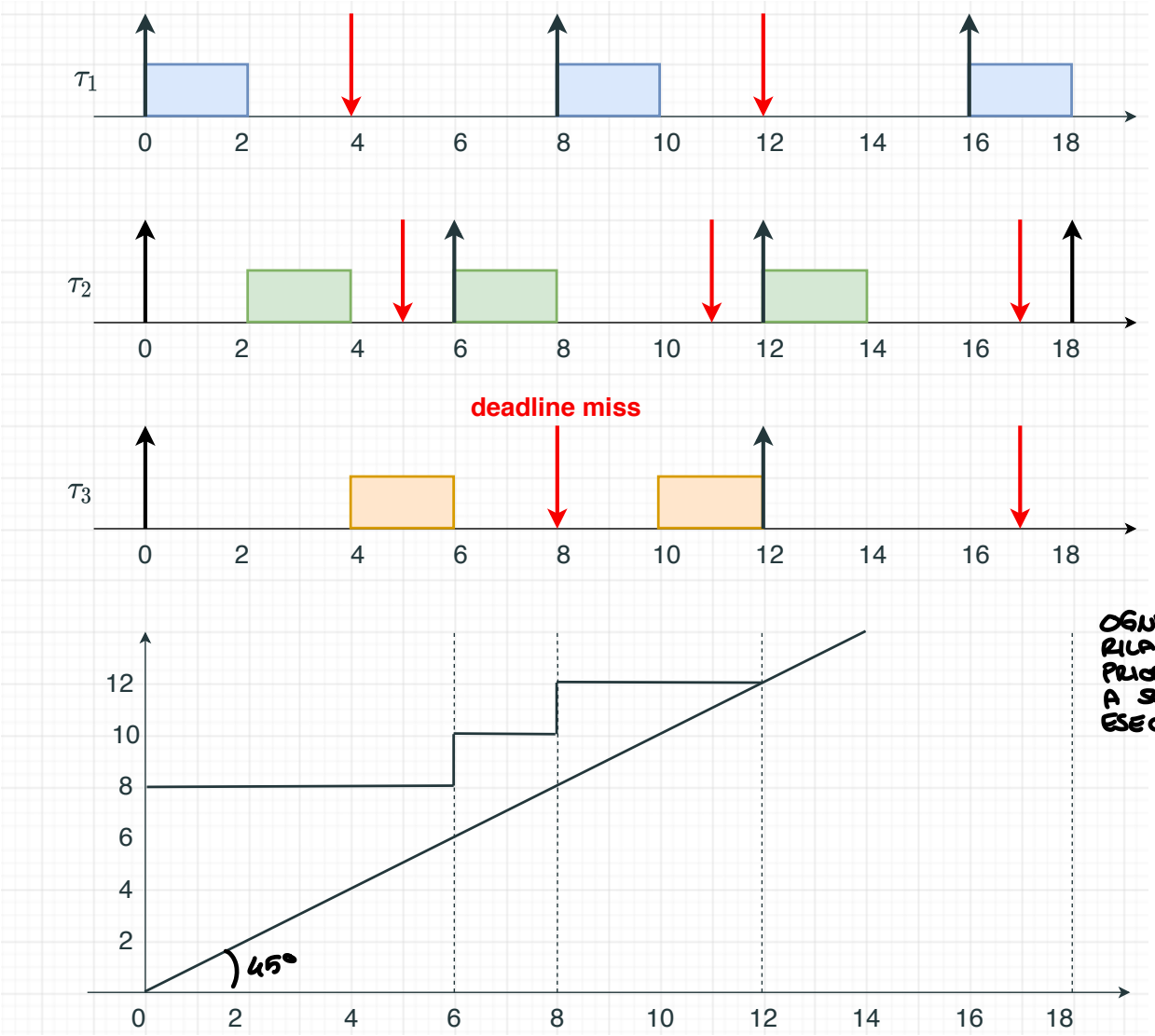


RESPONSE TIME ANALYSIS

- $R_1^{(0)} = C_1 = 2 < D_1$; $R_1^{(1)} = C_1 = 2 < D_1 \Rightarrow R_1 = 2$
- $R_2^{(0)} = C_1 + C_2 = 4 < D_2$; $R_2^{(1)} = C_2 + \lceil R_2^{(0)}/T_1 \rceil C_1 = C_2 + C_1 = 4 < D_2 \Rightarrow R_2 = 4$
  (note that the task response is the maximum among the job response times)
- $R_3^{(0)} = C_1 + C_2 + C_3 = 8 = D_3$; $R_3^{(1)} = C_3 + \lceil R_3^{(0)}/T_1 \rceil C_1 + \lceil R_3^{(0)}/T_2 \rceil C_2 =$
  $= C_3 + C_1 + 2C_2 = 10 > D_3 \Rightarrow$ the DM schedule is **unfeasible** for the task set

DM: example of an unfeasible schedule (2/2)

- The estimated response time increases at each task release: $R_3 = 12$



OGNI VOLTA CHE VIENE RILASCIATO UN TASK CON PRIORITÀ PIÙ ALTA LA FUNZIONE A SCALINI SALE DEL TEMPO DI ESECUZIONE DEL TASK.